

Pietro Di Nicco

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SUMMARY

Machine learning engineer and MSc graduate (110/110 with honors, top of class) specializing in deep learning, natural language processing, and large language models. Delivered 3 end-to-end AI projects achieving 74.3% accuracy on IMO-level math problems and 0.94 AUC on graph anomaly detection. Experienced in LLM fine-tuning, pretraining, quantization, and MLOps-oriented inference optimization using PyTorch, TensorFlow, and vLLM. Effective communicator able to present complex technical concepts clearly to both technical and non-technical audiences. Collaborative team player committed to knowledge sharing and clear technical documentation.

EDUCATION

MSc in Artificial Intelligence and Deep Learning

Sep 2024 - Jul 2026

University of Calabria | Final Grade: 110/110 with Honors

Thesis: Explored limitations of standard tokenization and developed DynamicFrequent-BPE, a novel tokenization algorithm designed to capture patterns in domain-specific languages underrepresented in training data.

BSc in Computer Engineering

Sep 2021 - Jul 2023

University of Calabria | Final Grade: 109/110

Thesis: Generative Pretrained Transformer in NLP.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, SQL, Haskell, Prolog

Machine Learning & Deep Learning:

Convolutional Neural Networks (CNNs), Reinforcement Learning, Object Detection, Diffusion Models, Clustering, Recurrent Neural Networks (LSTMs)

Natural Language Processing:

Large Language Models (LLMs), Fine-tuning, Pretraining, Quantization, DPO, PPO, GRPO, Tokenization, Transformer Architecture

Frameworks & Tools: PyTorch, TensorFlow, Keras, vLLM, Scikit-learn, OpenCV, Git, Docker, Hugging Face, Jupyter

MLOps & Engineering:

MLOps, Model inference optimization, CI/CD, Agile methodology, Tool Integrated Reasoning, Self-Consistency decoding

Other: Graph Neural Networks, Anomaly Detection, Computer Vision, Data Augmentation

PROJECTS

Math Problem Solving Pipeline using GPT-OSS-120b

Mar 2025 - Jun 2025

Technologies: Python, vLLM, Large Language Models, Tool Integrated Reasoning, Self-Consistency, Docker

- Designed and implemented an inference pipeline for solving hard mathematical problems using GPT-OSS-120b, an open-source 120B parameter model.
- Applied Tool Integrated Reasoning (TIR) with 8 parallel generations per problem, each augmented with Python code execution via a local Jupyter kernel.
- Achieved 74.3% accuracy (26 out of 35) on IMO-level competition problems without any fine-tuning, outperforming standard single-pass inference.

Full write-up: pietro-d.github.io/website.io/projects/mathLLM

Real-Time Ball Detection in Soccer Videos using YOLOv8

Oct 2024 - Dec 2024

Technologies: Python, YOLOv8, Object Detection, Computer Vision, OpenCV, PyTorch

- Fine-tuned YOLOv8-nano (3.2M parameters) for real-time soccer ball tracking at 100+ FPS on resource-constrained hardware.
- Designed a temporal train/validation split across 4 videos (3,693 training / 722 validation frames) to prevent future data leakage.
- Achieved MSE of 0.03 on held-out test videos using mosaic augmentation and composite loss (bounding box, classification, distribution focal loss).

Full write-up: pietro-d.github.io/website.io/projects/balldetection

Anomalous Node Detection using RAND

Jan 2025 - Mar 2025

Technologies: Python, Reinforcement Learning, Graph Neural Networks, Anomaly Detection, PyTorch

- Implemented RAND (Reinforcement learning-based Anomaly detection on graph Networks), an unsupervised RL framework for graph anomaly detection.
- Applied to AmazonRatingsDataset (24,492 nodes, 190,300 edges); achieved 0.94 AUC on combined topological and contextual anomaly detection.
- Conducted systematic hyperparameter analysis and tested alternative neighborhood strategies (Katz Centrality, Jaccard similarity), confirming model robustness.

Full write-up: pietro-d.github.io/website.io/projects/rand

LANGUAGES

Italian: Native | English: C1 (Proficient) - Listening C1, Reading C1, Speaking B2, Writing B2
